

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 December 2024 – January 2025

22MCA104 – MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	With example define the following terms. i) Singleton set ii) Equal set iii) Power set iv) Cardinality set v) Equivalent set	10	L*1	1	1
b)	Use set builder notation to give description of each of below set i) $\{-3, -2, -1, 0, 1, 2, 3\}$ ii) $\{7, 14, 21, 28\}$	4	L2	1	1
c)	Prove De Morgan's law (any one) using set builder notation and truth table.	6	L3		2
2. a)	Let A be a set of students who knows to dance and let B be set of students who knows to sing. Describe below expression in an English statement i) $A-B$ ii) $A \cup B$ iii) $B-A$	6	L2	1	1
b)	Show that a set of odd positive integers is a countable set.	4	L3	1	2
c)	Find the eigen values and eigen vectors of the following matrix. $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$	6	L3	1	2
d)	There are 20000 people living in Defense Colony. Out of them 9000 subscribe to Star TV Network and 12000 to Zee TV Network. If 4000 subscribe to both, how many do not Subscribe to any of the two?	4	L3	1	2
	Unit – II				
3. a)	$((\sim q \wedge (p \rightarrow q)) \rightarrow \sim p)$ is tautology using series of logical equivalence and truth table.	10	L3	2	2
b)	What are different types of quantifiers. Explain with example.	6	L2	2	1
c)	Differentiate between Conjunction and Disjunction.	4	L2	2	1
4. a)	Use Demorgan's law to find negation of "May your day be filled with unlimited blessing and with endless opportunities"	4	L3	2	2
b)	Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r))$ from input bits p, q, and r.	6	L2	2	1
c)	Represent the statement "Ravi is good in logic when Ravi is good in math" in Inverse, Contrapositive and Converse form.	6	L2	2	1
d)	Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent by developing a series of logical equivalence and truth table.	4	L3	2	2
	Unit – III				
5. a)	Consider the following relation on $\{1, 2, 3, 4\}$ $R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$, $R_2 = \{(1, 1), (1, 2), (2, 1)\}$, $R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$, $R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$, $R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$, $R_6 = \{(3, 4)\}$. Which of these relations are i.) Reflexive ii.) Symmetric iii.) Transitive iv) Anti symmetric	12	L2	3	1
b)	Find the transitive closure using Warshalls algorithm for the set $\{(2,1) (2,3) (3,1) (3,4) (4,1) (4,3)\}$	8	L3	3	2

P.T.O.

6. a) Find the smallest relation containing the relation $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ that is reflexive and transitive.
- b) Let R be the relation on set of integers such that aRb if and only if $a-b$ is an integer. Is R an equivalence relation?

10	L2	3	1
10	L3	3	2

Unit – IV

7. a) The marks obtained by 1000 students in an examination are found to be normally distributed with mean 70 and standard deviation 5. Estimate the number of students whose marks will be i) less than 65 ii) between 60 to 75, iii) more than 75
- b) Prove Bayes theorem.
- c) Given a discrete probability distribution. Find the value of K .

X	0.5	1	1.5	2
$P(x)$	K	$2K$	$3K$	K

10	L3	4	2
6	L2	4	1
4	L2	4	1

8. a) Given standard normal distribution, find the area under curve that lies
- i) To the right of $Z = 1.84$ and
- ii) Between $Z = -1.97$ and $Z = 0.86$

10	L2	4	1
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- b) Find the expected value of given discrete probability distribution

X	0	1	2	3	4
$P(x)$	0.44	0.36	0.15	0.04	0.01

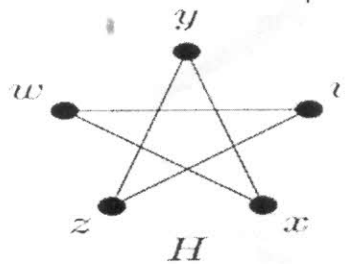
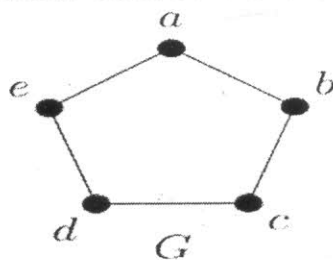
6	L2	4	1
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- c) A Die is rolled once. Find the probability of getting
- i) a number greater than 6
- ii) a number between 1 and 6

4	L2	4	1
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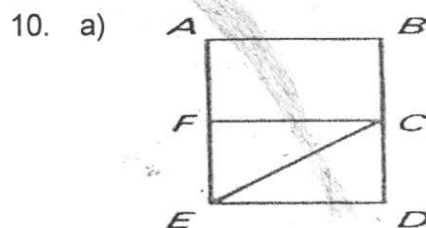
Unit – V

9. a) Check whether the two graphs given below are isomorphic or not



12	L3	5	2
8	L3	5	2

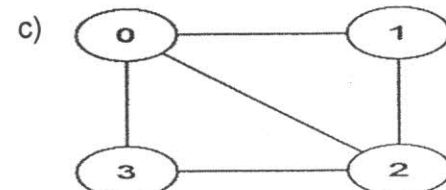
- b) Prove that number of vertices of odd degree is even.



Check whether the graph given above is Euler path or Euler circuit or both?

5	L3	5	2
5	L2	5	1

- b) Write a note on Konigsberg bridge problem.



Check whether the given graph is bipartite or not.

5	L3	5	2
5	L3	5	2

- d) Prove that a tree with n vertices has $n-1$ edges.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
